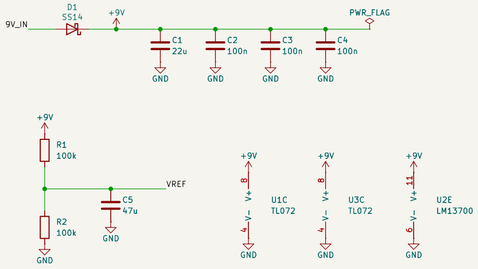
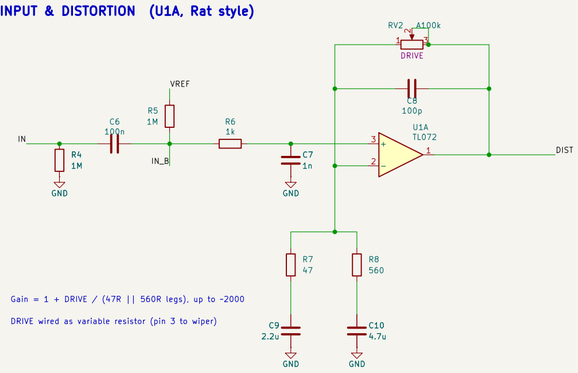
POWER & VREF



VREF = 4.5V, filtered (new)

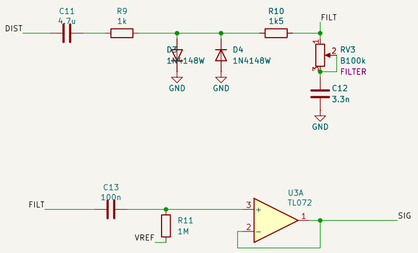
INPUT & DISTORTION (U1A, Rat style)



Gain = 1 + DRIVE / (47R || 560R legs), up to ~2000

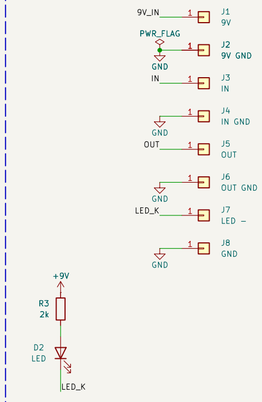
DRIVE wired as variable resistor (pin 3 to wiper)

CLIPPER, FILTER & BUFFER (U3A)

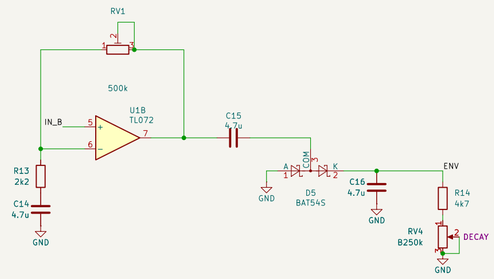


FILTER: CW = darker

OFF-BOARD WIRING



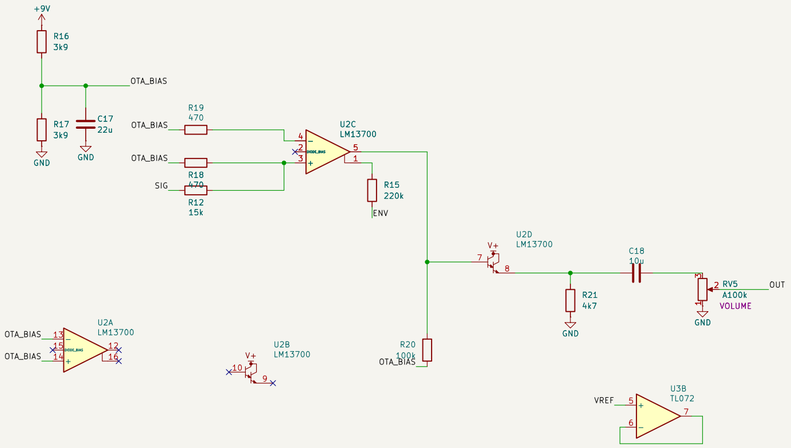
ENVELOPE FOLLOWER (U1B)



Release ~ C16 x (R15 || R14+DECAY): ~20 ms to ~200 ms

D5 = BAT54S: pin1 A to GND, pin3 common, pin2 K to hold cap

VCA (U2 LM13700) & OUTPUT



Unused halves: U2B inputs to OTA_BIAS, U3B as follower

NOTES

1. After R.A. Penfold "Fuzz Unit" (Music Projects, 1994); distortion re-imposed with the input envelope by an OTA VCA.
2. Changes: Rat-style gain stage with diodes to ground + FILTER; buffer U3A; DECAY pot; BAT54S detector; 1M input / filtered VREF; VOLUME; SENS trimmer.
3. SMD passives 0805 hand-solder pads (C5 1206, C12 0603), LCSC field = JLCPCB part number.
4. Hand-soldered: RV2–RV5 (Alpha 16mm pots), RV1 (3296W trimmer), D2 (3mm LED), J1–J8 wire pads.
5. Pots: VOLUME A100k, DRIVE A100k, FILTER B100k, DECAY B1M.

Enclosure: Hammond 1590B
Rat-style distortion into an envelope-controlled OTA VCA

Sheet: /
File: Penfold_Fuzz.kicad_sch

Title: Penfold Dynamic Fuzz – SMD

Size: A2	Date: 2026-09-25	**Rev: 1.0**
KiCad E.D.A. 10.0.4		Id: 1/1